Machine Learning In Image Analysis PS1

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1 Question 1

Prove the following properties of convolution. For all continuous functions f, g, and h, the following axioms hold. **Make sure to lay out all key steps of your proof for full credit**.

- **Associativity:** $(f * g) * h = f * (g * h)$
- **Distributivity:** $f * (g + h) = f * g + f * h$
- **Differentiation Rule:** $(f * g)' = f' * g = f * g'$
- **Convolution Theorem:** $\mathcal{F}(g * h) = \mathcal{F}(g)\mathcal{F}(h)$, where $\mathcal{F}$ denotes the Fourier transform.

Solution

Please note that the functions and notations used here are slightly different to as explained in class, However they are accurate and true.

Let f, g, and h be continuous functions. The convolution of two functions is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)\, d\tau.$$

We will prove the following properties using the integral definition of convolution.

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1. Associativity

Statement: $(f * g) * h = f * (g * h)$.

Proof:

First, compute $(f * g) * h$:

$$\begin{aligned}((f * g) * h)(t) &= \int_{-\infty}^{\infty} (f * g)(\tau)h(t - \tau)\, d\tau \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(u)g(\tau - u)\, du\right] h(t - \tau)\, d\tau.\end{aligned}$$

Switch the order of integration (by Fubini's theorem):

$$((f * g) * h)(t) = \int_{-\infty}^{\infty} f(u) \left[\int_{-\infty}^{\infty} g(\tau - u) h(t - \tau) \, d\tau \right] du.$$

Let $s = \tau - u$, so $\tau = s + u$ and $d\tau = ds$. Then,

$$\begin{aligned} ((f * g) * h)(t) &= \int_{-\infty}^{\infty} f(u) \left[\int_{-\infty}^{\infty} g(s) h(t - s - u) \, ds \right] du \\ &= \int_{-\infty}^{\infty} f(u) \left[(g * h)(t - u) \right] du \\ &= (f * (g * h))(t). \end{aligned}$$

Conclusion: $(f * g) * h = f * (g * h)$.
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2. Distributivity

Statement: $f * (g + h) = f * g + f * h$.
Proof:
Compute $f * (g + h)$:

$$\begin{aligned} (f * (g + h))(t) &= \int_{-\infty}^{\infty} f(\tau) [g + h](t - \tau) \, d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) [g(t - \tau) + h(t - \tau)] \, d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) g(t - \tau) \, d\tau + \int_{-\infty}^{\infty} f(\tau) h(t - \tau) \, d\tau \\ &= (f * g)(t) + (f * h)(t). \end{aligned}$$

Conclusion: $f * (g + h) = f * g + f * h$.
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3. Differentiation Rule

Statement: $(f * g)' = f' * g = f * g'$.
Proof:
First Part: $(f * g)' = f * g'$
Compute the derivative of the convolution:

$$\begin{aligned} (f * g)'(t) &= \frac{d}{dt} \int_{-\infty}^{\infty} f(\tau) g(t - \tau) \, d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) \frac{d}{dt} g(t - \tau) \, d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) g'(t - \tau) \, d\tau \\ &= (f * g')(t). \end{aligned}$$

Second Part: $(f * g)' = f' * g$

Alternatively, consider:

$$\begin{aligned}
(f * g)(t) &= \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau \\
&= \int_{-\infty}^{\infty} f(t-s)g(s)\,ds \quad (\text{Let } s = t-\tau) \\
&= \int_{-\infty}^{\infty} f(t-s)g(s)\,ds.
\end{aligned}$$

Differentiate with respect to t:

$$\begin{aligned}
(f * g)'(t) &= \int_{-\infty}^{\infty} \frac{d}{dt} f(t-s)g(s)\,ds \\
&= \int_{-\infty}^{\infty} f'(t-s)g(s)\,ds \\
&= (f' * g)(t).
\end{aligned}$$

Conclusion: $(f * g)' = f' * g = f * g'$.
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4. Convolution Theorem

Statement: $\mathcal{F}(g * h) = \mathcal{F}(g) \cdot \mathcal{F}(h)$, where $\mathcal{F}$ denotes the Fourier transform.
Proof:
definition of the Fourier transform:

$$\mathcal{F}(f)(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,dt.$$

Compute $\mathcal{F}(g * h)(\omega)$:

$$\begin{aligned}
\mathcal{F}(g * h)(\omega) &= \int_{-\infty}^{\infty} (g * h)(t)e^{-i\omega t}\,dt \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} g(\tau)h(t-\tau)\,d\tau\right] e^{-i\omega t}\,dt \\
&= \int_{-\infty}^{\infty} g(\tau) \left[\int_{-\infty}^{\infty} h(t-\tau)e^{-i\omega t}\,dt\right] d\tau.
\end{aligned}$$

Let $s = t - \tau$, so $t = s + \tau$ and $dt = ds$:

$$\begin{aligned}
\mathcal{F}(g * h)(\omega) &= \int_{-\infty}^{\infty} g(\tau) \left[\int_{-\infty}^{\infty} h(s)e^{-i\omega(s+\tau)}\,ds\right] d\tau \\
&= \int_{-\infty}^{\infty} g(\tau)e^{-i\omega\tau} \left[\int_{-\infty}^{\infty} h(s)e^{-i\omega s}\,ds\right] d\tau \\
&= \left[\int_{-\infty}^{\infty} g(\tau)e^{-i\omega\tau}\,d\tau\right] \left[\int_{-\infty}^{\infty} h(s)e^{-i\omega s}\,ds\right] \\
&= \mathcal{F}(g)(\omega) \cdot \mathcal{F}(h)(\omega).
\end{aligned}$$

Conclusion: $\mathcal{F}(g * h) = \mathcal{F}(g) \cdot \mathcal{F}(h)$.

2 Question 2

Frequency Smoothing

(a) Compute the Fourier transform of the given image `lenaNoise.PNG` by using `numpy.fft.fft2` in Python, and then center the low frequencies (e.g., by using `fftshift`).

(b) Keep a different number of low frequencies (e.g., 72, 152, 312, and the full dimension), but set all other high frequencies to 0.

(c) Reconstruct the original image (using `ifft2`) by using the new generated frequencies in step (b).

Solution

Solution In Jupyter Notebook which is Also Attached After The Report

3 Question 3

3.1 Problem Description

Image denoising is the process of removing noise from a given noisy image to recover a clean image. Let $f \in \mathbb{R}^{m\times n}$ be the observed noisy image, where each pixel value contains additive noise. The goal is to find a clean image $u \in \mathbb{R}^{m\times n}$ that is as close as possible to the noisy image f, while ensuring that the resulting image u is smooth and free from noise.

One common approach to achieve this is by minimizing an energy function that consists of two terms:

- A data fidelity term, which ensures that u stays close to the noisy image f.
- A regularization term, which imposes smoothness on u while preserving important features like edges.

3.2 Objective Function

The objective function to minimize for image denoising is the Total Variation (TV) [1] regularized energy function:

$$E(u) = \lambda \|f - u\|_{L^2}^2 + \int_\Omega \|\nabla u(x)\|\, dx$$

where:

- $\|f-u\|_{L^2}^2 = \int_\Omega (f(x)-u(x))^2\, dx$ is the data fidelity term that measures the difference between the noisy image f and the clean image u.
- $\int_\Omega \|\nabla u(x)\|\, dx$ is the Total Variation (TV) norm, which enforces smoothness on the image by penalizing large gradients in u.

- $\lambda > 0$ is a regularization parameter that controls the trade-off between the data fidelity and smoothness.

In a discrete setting, the objective function becomes:

$$E(u) = \lambda \sum_{i,j} (f_{i,j} - u_{i,j})^2 + \sum_{i,j} \sqrt{(D_x u_{i,j})^2 + (D_y u_{i,j})^2}$$

where $D_x u_{i,j}$ and $D_y u_{i,j}$ are the discrete gradients in the x and y directions.

3.3 Gradient of the Objective Function

The gradient of the objective function $E(u)$ with respect to u is given by:

$$\nabla_u E(u) = 2\lambda(u - f) - \mathrm{div}\left(\frac{\nabla u}{\|\nabla u\|}\right)$$

where:

- $2\lambda(u - f)$ is the gradient of the data fidelity term.
- $\mathrm{div}\left(\frac{\nabla u}{\|\nabla u\|}\right)$ is the gradient of the Total Variation (TV) regularization term.
- $\|\nabla u\| = \sqrt{(D_x u)^2 + (D_y u)^2}$ is the magnitude of the image gradient.

3.4 Gradient Descent Algorithm

To minimize the objective function $E(u)$, we use the gradient descent algorithm. The update rule at iteration k is:

$$u^{k+1} = u^k - \alpha \nabla_u E(u^k)$$

where:

- $\alpha > 0$ is the learning rate that controls the step size of each update.
- $\nabla_u E(u^k)$ is the gradient of the objective function evaluated at u^k.

At each iteration, the clean image u is updated by subtracting the gradient of the objective function scaled by the learning rate α.

3.5 Results And Discussion

Convergence Graphs, Resulting Images After Denoising and Discussion are In The Jupyter Notebook which is also attached after the report.

References

[1] L. I. Rudin, S. Osher, E. Fatemi, *Nonlinear total variation based noise removal algorithms*, Physica D: Nonlinear Phenomena, Volume 60, Issue 1-4, 1992, Pages 259-268.

[2] OpenAI, ChatGPT, *ChatGPT: Language Model for Conversations*, OpenAI, 2023. Available at: `https://openai.com/chatgpt`.